Seeking Alpha Through Optimal Asset Allocation Across Market Environments and Alternative Data

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Overview

- Determine the optimal asset allocation across equities, bonds, gold, and real estate under four primary market environments:

Rising Growth/ Rising Inflation	**Rising Growth / Falling Inflation**
Falling Growth/ Rising Inflation	**Falling Growth/ Falling Inflation**

- Evaluate which asset classes tend to perform best within each of the four market environments. This analysis examines the performance of five asset classes over the period 1976–2025: Equities, Bonds, Real Estate (House Price Index), and Gold.
- Develop signals derived from sensitive market data to identify impending shifts in market environments, enabling dynamic asset allocation that preserves capital while capturing growth opportunities. (IN PROGRESS)

Tractable Data (1976 – 2025)

- **Equities:** S&P 500 index to measure stock market performance.
- **Bonds:** U.S. Treasury yields (2-year and 10-year) to approximate bond returns.
- **Commodities:** Gold prices (GLD) to capture commodity performance
- **Real Estate:** House Price Index
- **Macroeconomic Indicators:**
- **Inflation:** CPI (Consumer Price Index) for measuring price changes.
- **Economic growth:** Industrial Production Index (INDPRO) as a proxy for GDP growth.

Data Retrieval

- Equities: Investing.com
- US Treasury Yields (2-year and 10-year bonds), All-Transactions House Price Index for the US, CPI, INDPRO: Federal Reserve Economic Data (FRED) - using API key
- Gold (GLD): Macrotrend

Data Analysis

Year	Inflation_Growth	IP_Growth	SP500_Return	Gold_Return	RealEstate_Return	GS10_Bond_Return	GS2_Bond_Return	Regimes_Zero_Threshold
1976	5.20	5.83	1.09	3.20	10.48	11.72		Rising-Growth / Rising-Inflation
1977	6.81	5.14	-12.55	32.63	14.39	1.59	3.04	Rising-Growth / Rising-Inflation
1978	9.25	7.90	12.00	33.02	15.32	-0.59	3.22	Rising-Growth / Rising-Inflation
1979	13.87	0.85	14.31	186.63	9.43	-3.65	6.91	Rising-Growth / Rising-Inflation
1980	11.79	-1.97	13.49	-24.03	5.07	-2.47	8.33	Falling-Growth / Rising-Inflation
1981	8.26	-4.26	-7.10	-23.50	1.02	-2.58	10.90	Falling-Growth / Rising-Inflation
1982	3.71	-2.14	20.68	29.56	7.72	45.57	24.00	Falling-Growth / Rising-Inflation
1983	4.29	11.03	12.46	-26.06	4.33	1.39	6.97	Rising-Growth / Rising-Inflation
1984	3.53	2.89	9.91	-18.02	4.57	13.85	11.92	Rising-Growth / Rising-Inflation
1985	3.97	2.04	17.93	16.10	6.29	27.81	13.15	Rising-Growth / Rising-Inflation
1986	1.36	0.61	29.41	15.06	7.34	25.02	11.58	Rising-Growth / Rising-Inflation
1987	4.13	7.68	-6.20	14.23	5.00	-4.85	3.71	Rising-Growth / Rising-Inflation
1988	4.48	3.08	15.71	-15.13	5.25	5.52	4.84	Rising-Growth / Rising-Inflation
1989	5.20	-0.85	10.62	5.75	4.96	15.69	11.14	Falling-Growth / Rising-Inflation
1990	5.65	-0.96	4.50	-11.48	1.48	9.11	9.82	Falling-Growth / Rising-Inflation
1991	2.67	0.53	18.87	-4.18	3.06	16.04	11.04	Rising-Growth / Rising-Inflation
1992	3.25	5.10	7.34	-6.61	1.67	10.26	5.99	Rising-Growth / Rising-Inflation
1993	2.45	3.31	9.75	14.35	3.25	12.98	4.84	Rising-Growth / Rising-Inflation
1994	2.87	6.80	-2.33	-0.62	1.45	-9.48	-1.93	Rising-Growth / Rising-Inflation
1995	2.79	1.88	35.20	7.28	5.17	23.76	11.83	Rising-Growth / Rising-Inflation
1996	3.04	6.95	23.62	-14.43	2.23	-1.33	3.49	Rising-Growth / Rising-Inflation
1997	1.63	8.58	24.69	-12.52	5.09	14.38	7.18	Rising-Growth / Rising-Inflation
1998	1.67	3.76	30.53	-5.35	4.59	11.69	6.69	Rising-Growth / Rising-Inflation
1999	2.79	4.62	8.98	-0.91	5.81	-9.83	1.34	Rising-Growth / Rising-Inflation
2000	3.72	0.40	-2.04	-5.95	7.67	17.91	9.46	Rising-Growth / Rising-Inflation

Data Analysis

Year	Inflation_Growth	IP_Growth	SP500_Return	Gold_Return	RealEstate_Return	GS10_Bond_Return	GS2_Bond_Return	Regimes_Zero_Threshold
2000	3.72	0.40	-2.04	-5.95	7.67	17.91	9.46	Rising-Growth / Rising-Inflation
2001	1.20	-3.68	-17.26	5.75	6.02	6.06	7.87	Falling-Growth / Rising-Inflation
2002	2.76	3.07	-24.29	31.56	6.48	12.47	5.35	Rising-Growth / Rising-Inflation
2003	2.03	1.28	32.18	8.36	7.35	3.30	1.70	Rising-Growth / Rising-Inflation
2004	2.84	3.77	4.44	5.61	11.12	3.63	-0.87	Rising-Growth / Rising-Inflation
2005	4.02	2.42	8.36	34.45	10.46	2.72	1.10	Rising-Growth / Rising-Inflation
2006	2.08	1.66	12.35	13.33	3.29	1.87	3.54	Rising-Growth / Rising-Inflation
2007	4.29	2.41	-4.15	43.05	-2.23	12.41	9.20	Rising-Growth / Rising-Inflation
2008	-0.11	-13.37	-40.09	-0.57	-5.78	12.89	5.49	Falling-Growth / Falling-Inflation
2009	2.62	0.87	30.03	17.88	-7.01	-6.56	0.59	Rising-Growth / Rising-Inflation
2010	1.70	4.63	19.76	23.16	-3.42	6.28	1.51	Rising-Growth / Rising-Inflation
2011	3.01	3.69	2.04	30.33	-1.62	14.04	1.28	Rising-Growth / Rising-Inflation
2012	1.68	1.45	14.15	-3.65	2.16	2.42	0.19	Rising-Growth / Rising-Inflation
2013	1.56	1.86	18.99	-25.56	4.62	-5.22	0.05	Rising-Growth / Rising-Inflation
2014	-0.23	2.73	11.91	1.36	5.40	10.21	0.10	Rising-Growth / Falling-Inflation
2015	1.24	-3.24	-2.74	-11.64	4.84	0.30	-0.08	Falling-Growth / Rising-Inflation
2016	2.51	-0.80	17.45	8.51	5.22	-0.46	0.34	Falling-Growth / Rising-Inflation
2017	2.15	2.74	23.91	10.89	6.14	1.31	-0.27	Rising-Growth / Rising-Inflation
2018	1.49	1.91	-4.24	-1.55	4.59	1.61	1.11	Rising-Growth / Rising-Inflation
2019	2.60	-2.29	19.28	19.53	5.09	9.84	4.38	Falling-Growth / Rising-Inflation
2020	1.37	-2.13	15.15	17.01	7.59	6.86	4.02	Falling-Growth / Rising-Inflation
2021	7.56	1.32	21.57	-2.90	19.43	-4.02	-1.40	Rising-Growth / Rising-Inflation
2022	6.33	0.32	-9.72	7.35	7.63	-11.52	-4.83	Rising-Growth / Rising-Inflation
2023	3.09	-1.28	18.86	5.76	6.25	-0.45	4.01	Falling-Growth / Rising-Inflation
2024	2.99	0.85	24.66	37.22	4.86	-0.22	4.41	Rising-Growth / Rising-Inflation
2025	2.39	2.33	14.87	73.87		7.78	5.58	Rising-Growth / Rising-Inflation

Key Findings

1. **Rising-Growth / Rising-Inflation (The Expansion)**

Occurred 37 times (74% of the time). This is the "Standard" state of the U.S. economy.

Asset	Avg. Return	Win Rate
S&P 500	11.72%	78.4%
Gold	13.95%	59.5%
Real Estate	5.45%	88.9%

- **S&P 500:** Corporate earnings grow alongside the economy. While inflation rises, companies with "pricing power" can pass costs to consumers, keeping profit margins healthy.
- **Gold:** Investors buy gold as a hedge against the eroding purchasing power of the dollar that usually accompanies rising inflation.
- **Real Estate:** Higher demand for housing and the ability to raise rents during expansions make this the most consistent "win" in this regime.

Key Findings

2. Falling-Growth / Rising-Inflation (The Stagflation)

Occurred 11 times. This is a difficult environment where the economy slows but prices keep climbing.

Asset	Avg. Return	Win Rate
2-Year Bond	7.70%	90.9%
Real Estate	5.02%	100.0%
10-Year Bond	7.95%	63.6%

- **Real Estate:** It acts as a "hard asset" that retains value during inflation, while the essential nature of housing keeps demand from collapsing even when growth slows.
- **2-Year Bonds:** When growth falls, investors seek safety. Short-term bonds are less volatile than the 10-year and allowed investors in the 70s/80s to "re-invest" at higher interest rates as inflation stayed high.

Key Findings

3. Falling-Growth / Falling-Inflation (The Deflationary Bust)

Occurred in 2008 (Global Financial Crisis). This is the most dangerous regime for investors.

- **10-Year Bonds:** This is the "Ultimate Insurance." When growth and inflation both crash, the Federal Reserve aggressively cuts interest rates. Because bond prices move opposite to rates, long-term bonds see massive price surges.
- **Why Stocks fail:** Falling growth destroys earnings, and falling inflation suggests a lack of consumer demand, leading to a "liquidation" event in the stock market.

Key Findings

4. Rising-Growth / Falling-Inflation (The Goldilocks)

- Occurred in 1986. The "Best of all worlds" for financial markets.
- **S&P 500:** Growth is healthy for earnings, but the lack of inflation means the Fed doesn't need to raise rates. This creates "cheap money" for companies to expand.
- **10-Year Bonds:** As inflation falls, the "real yield" of existing bonds becomes more attractive, and the market anticipates lower future interest rates, driving bond prices up alongside stocks.

Implication for stakeholders

- Individual investors: Provides a defensive framework for capital growth while introducing a new approach to generating alpha.
- Fund Managers: Offering data-driven insights on when to pivot between macro- environments.

Ethical, legal, societal implications

- Serves as a reference for individual investors, without guaranteeing future results.
- This is for informational purposes only – we are not registered investment advisors, so please use with caution.

Engagement and originality

- Backtest the model using historical data to evaluate its performance and practical applicability.
- Next step: determine how to both preserve capital, and generate alpha within the portfolio.